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Strategies to Achieve the African Union's 10-Year Action Plan for Fertilizer and Soil Health

A Policy Brief

CGIAR

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Excellence in Agronomy

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Executive Summary

This policy brief addresses the challenges and strategies for achieving the African Union's 10-year action plan for fertilizer and soil health. Key recommendations include enhancing multi-sectoral partnerships, advancing research and development, fostering intentional investment, and promoting inclusive capacity development.

Introduction

Africa's agricultural sector is vital for the continent's food security and economic growth. However, soil degradation and low agricultural productivity pose significant challenges. This policy brief aims to provide strategic actions to address these issues and support the implementation of the African Union's 10-year action plan for fertilizer and soil health. It outlines practical strategies and recommendations for improving soil health and fertilizer use in Africa, thereby enhancing agricultural productivity and sustainability.

Status Quo and Challenges of African Agriculture

Africa's vast arable land, which covers about 200 million hectares, holds significant agricultural potential. However, the continent faces substantial challenges in agricultural productivity, including climate change impacts, soil degradation, and socio-economic and institutional barriers. The projected population growth from 1.4 billion in 2023 to approximately 2.5 billion by 2050 will exacerbate these challenges. Addressing these issues requires comprehensive, multi-actor interventions to enhance agricultural productivity and ensure food security across the continent.

Enhancement of agricultural productivity depends critically on restoring and sustainably managing soil health. However, soil health and agricultural productivity are at severe risk due to continuing land degradation, especially in smallholder farming areas. These areas face persistent food shortages due to challenging agro-ecological and socio-economic conditions. This situation has been aggravated by the fact that fertilizer use has not reached its full potential. Many African countries still deal with declining fertilizer use per hectare, resulting from poor soil management practices, lack of access to quality fertilizers, insufficient investment in agricultural research and development, and limited capacity among farmers to adopt best practices. Only some countries have achieved the Abuja target of 50 kg ha^{-1} fertilizer use. From 2006 to 2022, fertilizer use increased in average from 8 to 18 kg ha^{-1} of cropland, comprising 13% of the global average (135 kg ha^{-1}) (African Union, 2024). Hence, the continent needs sustainable and scalable agronomic solutions to increase fertilizer use while benefiting soil health, the backbone of sustainable crop production.

Soil Health and Agronomy

Soil health is fundamental to agriculture and environmental well-being. It refers to the soil's capacity to function as a living system, supporting plant and animal productivity, maintaining water and air quality, and promoting ecosystem health. Soil health is crucial for food security, biodiversity conservation, and climate change mitigation. In agronomic terms, it encompasses the physical, chemical, and biological properties that sustain crops and livestock. Maintaining and improving soil health is vital for sustainable and profitable crop yields, pest and disease resistance, efficient input use, and resilience to climate change.

Agronomy plays a key role in creating productive and sustainable agricultural systems that support long-term food security, soil health, and environmental welfare. Agronomists are responsible for developing and scaling practices that enhance soil health, such as integrated soil fertility management (ISFM). ISFM involves using a combination of fertilizers, organic inputs, and improved germplasm, tailored to local conditions, to enhance nutrient efficiency and crop productivity while adhering to sound agronomic practices (Vanlauwe et al., 2010). Establishing a positive relationship between agronomic practices and soil health is essential for improving agricultural profitability, food security, and sustainable development while ensuring environmental safety.

Key Findings	
1	**Agricultural Transformation:** Necessary for achieving food security, economic development and sustainability across Africa.
2	**Soil Health Importance:** Healthy soils are crucial for increasing agricultural productivity and maintaining environmental balance and enhancing resilience to climate change.
3	**Barriers to Adoption:** Limited access to resources, knowledge gaps, and inadequate infrastructure hinder the adoption of sustainable agronomic practices benefiting soil health.

Recommendations

Addressing challenges in soil health and fertilizer use requires coordinated efforts, innovative solutions, and substantial investment in research and development. To drive impactful change, actionable and high-impact leverage points must be implemented as part of the African Union's 10-year action plan for fertilizer and soil health endorsed at the Africa Fertilizer and Soil Health Summit in Nairobi in May 2024. The summit highlighted the necessity of increasing fertilizer use in Africa while supporting soil health and environmental resilience. Enhancing the efficiency and effectiveness of inputs is crucial for maximizing agricultural productivity and profitability, soil health and biodiversity conservation, as well as climate change resilience and reducing agriculture's footprint on nature.

Persistent operational hurdles need to be addressed for deploying the proposed leverage points. Challenges related to soil health and efficient fertilizer use require further exploration, understanding, and coordination. Efforts have often been reactive or focused on short-term goals, lacking integrative frameworks for designing, implementing, and evaluating solutions across different initiatives, sectors, and scales. Consequently, only a small yet influential group of stakeholders, including farmers and

policymakers, have adopted innovative soil health practices.

Implementing the proposed leverage points will maximize impact and provide pathways for enhancing soil health, advancing agronomy, industry, and policy progress. These actions offer a practical, coordinated, and inclusive guide for executing the African Union's 10-year action plan for fertilizer and soil health, aiding stakeholders in prioritizing and coordinating actions, allocating resources, monitoring progress, and maximizing impact over time. The suggested recommendations bridge the gap between theory and practice, promoting soil health for improved agricultural productivity, environmental resilience, sustainable development, and inclusivity.

High-Impact Leverage Points		
1		**Trusted multi-sectoral partnerships** Encouraging the adoption of soil health-conserving agronomic practices, especially ISFM, is crucial but challenging. Technological innovations can significantly advance these practices by enhancing soil health and optimizing fertilizer use efficiency. To ensure widespread adoption and practical outcomes, effective integration of these innovations with scalable efforts is essential. This requires coordinated efforts between the public and private sectors to optimize resource use and comprehensively support farmers. Establishing iterative feedback systems is vital for continuously improving agronomic practices, allowing stakeholders to monitor effectiveness and adjust strategies as needed. Such trusted multi-sector partnerships and feedback mechanisms aim to create a resilient and productive agricultural sector that addresses evolving environmental and socio-economic needs, ensuring sustainable and inclusive solutions for all stakeholders.
2		**Demand-driven research & development (R&D)** Advancing soil health in Africa's agronomy requires local, technical solutions supported by R&D. Effective agronomic practices and monitoring tools are essential for improving soil health and optimizing fertilizer use. The lack of affordable, accurate on-field soil health measurement tools limits the adoption of sustainable practices and timely farmer responses. Integrating soil health data into accessible digital platforms is crucial for informed decision-making. R&D efforts focusing on advanced technologies like remote sensing and digital advisory tools can enhance precision agriculture and climate-resilient farming. Investing in technological innovation and stakeholder training is key to bridging the gap between research and practical application, fostering a resilient agricultural sector in Africa.
3		**Intentional investment and finance** Soil health is becoming increasingly important for investors in agriculture due to its significant impact on business sustainability. Public-private collaborations are essential for promoting economically viable soil health management practices and scaling sustainable agriculture. Increased private sector investment and targeted R&D are crucial for developing inclusive agronomic practices and investment models that optimize fertilizer use while protecting soil health. Prioritizing agricultural practices that balance immediate needs with long-term sustainability is critical. Tailored incentives and support mechanisms are needed to encourage the adoption of soil health-promoting practices like ISFM. Investments in soil health research, infrastructure, and capacity building drive agronomic innovation. Smallholder farmers, facing financial constraints, require support from governments and private sectors. Improving lease arrangements to incentivize long-term soil health improvements is also crucial. Agricultural retailers can facilitate the dissemination of agronomic knowledge and adapt to soil health innovations. Coordinated efforts and sustained investment will enhance agricultural resilience, providing lasting environmental and economic benefits.
4		**Inclusive capacity development** Farmers need localized evidence and tailored strategies to effectively integrate soil health practices on their land. Developing local expertise and capacity is crucial for scaling sustainable agronomy practices across diverse agricultural contexts, including gender-specific needs. This involves educating farmers, including women and marginalized groups, on operational strategies and regional variabilities to optimize the adoption of inclusive practices. Policy initiatives should prioritize building these skills among all farming communities and stakeholders. Co-creation, awareness campaigns, educational programs, farmer-to-farmer exchange models, and extension services are essential for promoting soil health management and efficient fertilizer use. Enhanced understanding of local soil conditions enables informed decision-making, reduces costs, and boosts profitability. Efforts should ensure that women farmers and other vulnerable groups have equal access to new agronomic technologies and innovations. R&D plays a vital role in addressing capacity gaps and developing effective agronomic solutions. Empowering extension agents, including women, through training and tools is crucial for translating research into practical applications, fostering resilient and prosperous farming communities.

Excellence in Agronomy Initiative and its Success Stories

The Excellence in Agronomy (EiA) Initiative of the CGIAR addresses urgent fertilizer and soil health challenges discussed in this policy brief. As a global accelerator, the EiA initiative develops scalable, evidence-based, and inclusive agronomic solutions and empowers farmers with essential tools and knowledge needed to revitalize and sustain their land. Aiming to enhance agricultural productivity for millions of smallholder farms by 2030, focusing on women and young farmers, the EiA Initiative seeks to improve food security, income, resource use, soil health, and climate resilience. The EiA Initiative promotes agronomy as the key to achieving soil health and efficient fertilizer use. To unlock the full potential of soil health, EiA supports smallholder farmers in achieving:

- Increased yields and profitability per unit of land (agronomic gain): Prioritizing sustainable intensification of existing farmland, emphasizing the minimization of deforestation and land conversion for agricultural purposes.
- Enhanced resource use efficiency per unit of investment: Optimizing nutrient and water utilization to minimize environmental impact, while enhancing labour efficiency.
- Enhanced climate adaptation, yield stability and reduced risk potential: Enhancing soil management strategies for increased resilience against climate shocks, including drought, heat, and flooding.
- Empowered livelihoods: Promoting women and youth in farming through inclusive knowledge, technology, investment and land access.

The selected success stories of the EiA initiative demonstrate the effectiveness of sustainable soil health initiatives and innovative fertilizer practices. With this, the EiA initiative drives significant progress towards achieving the African Union's 10-year action plan for fertilizer and soil health.

Ethiopia

Challenge: Declining soil fertility is a major cause of low crop productivity in Ethiopia. Many Ethiopian farming systems have negative nutrient balances, indicating that farmers are depleting soil nutrients, leading to sub-optimal yields and food insecurity. Deficiencies in micro and secondary nutrients limit the production potential of both rainfed and irrigated agriculture. To meet the food needs of a growing population without encroaching into forested areas or causing environmental degradation, agricultural intensification through efficient use of fertilizers and organic resources is required.

Solution: The EiA Initiative in cooperation with external and local partners has created an app-based digital decision support tool (DST) for site-specific fertilizer recommendations. It includes district maps and uses advanced analytics to provide valuable advice for wheat, teff, and sorghum systems. The tool supports national soil health investments and the Ministry of Agriculture's strategy, reaching 500 extension agents and 20,000 farmers across various regions including Amhara, Tigray, SNNP, and Oromia.

Gender Integration: The DST integrates user-centred design principles, creating versions tailored for different users. Recommendation domains consider socio-economic. scenarios to ensure female-led households and highly vulnerable groups are reached.

Pathway to Impact: Farmers using the DST will realize at least a 30% increase in yields due to better fertilizer targeting and balanced application. Farmers will access information and gain knowledge through the DST, enabling them to implement better farming practices. It is the ultimate goal to scale the developed DST to 5 million farmers by 2030, prioritizing fertilizer investments in the most responsive landscapes and farming systems. Read more

Rwanda

Challenge: Rwanda's major crop yields fall significantly short of their potential due to inadequate crop and nutrient management practices. The widespread use of uniform fertilizer recommendations leads to inefficient resource utilization and low profit margins. Site-specific fertilizer guidance could boost yields, reduce environmental risks by optimizing fertilizer use, and enhance returns on investments in the Ministry of Agriculture's (MINAGRI) fertilizer subsidy program under the Crop Intensification Program. There is a need for digital agricultural advisory solutions to provide customized agronomic guidance to farmers, with MINAGRI emphasizing the importance of digital tools for agricultural extension, especially for personalized fertilizer recommendations.

Solution: The EiA Initiative, in collaboration with the Rwanda Agriculture and Animal Resources Board (RAB) and the Rwanda Soil Information System (RwaSIS) project, has developed a fertilizer recommendation tool for six key crops (cassava, maize, wheat, potato, rice, beans). This tool integrates with the Smart Nkunganire System (SNS), a digital platform with over 1.5 million registered users, used for agro-input supply management. The tool provides location-specific fertilizer recommendations and serves as a model for integrating other forms of agronomic advice into the SNS framework.

Gender Integration: Female farmers are frequently underrepresented in the development and dissemination of advisory tools, resulting in gender disparities in productivity and access to technology. To address this, our local and external partners prioritize the inclusion of female farmers' needs in the design and delivery of these tools, ensuring equitable access and support.

Pathway to Impact: The initiative aims to boost crop yields by at least 30% through tailored fertilizer recommendations. It will pilot the tool with over 10,000 smallholder farmers by the end of 2025, refining recommendations and measuring their impact. If successful, the initiative will expand to reach hundreds of thousands of farmers in collaboration with MINAGRI and other stakeholders. The ultimate goal is to benefit over 1,000,000 smallholder farmers across multiple crops, significantly enhancing crop yields and food security in Rwanda. Read more

Nigeria

Challenge: Declining soil fertility is a major factor limiting crop yields for most farmers in Nigeria. Farmers often use blanket fertilizer recommendations for crops like maize, rice, and cassava, which do not account for soil fertility variability and crop management differences between fields. This results in nutrient imbalances and low fertilizer use efficiency. Recently, field-specific fertilizer recommendation tools for maize (Nutrient Expert), cassava (AKILIMO), and rice (RiceAdvice) have been developed and validated in Nigeria. These tools, created with input from R&D, fertilizer industry, and extension services, will significantly improve productivity, profitability, and fertilizer use efficiency if effectively disseminated to farmers.

Solution: The EiA initiative in cooperation with external and local partners develops and validates a digital agricultural advisory service to help extension agents provide integrated fertilizer recommendations and relevant agronomic advice. The service supports maize, rice, and cassava growers with a digital tool that offers tailored recommendations and investment prioritization advice for the three crops.

Gender Integration: Female farmers are often overlooked in the development and dissemination of advisory tools, leading to gender inequalities in productivity and technology access. The Sasakawa Africa Association (SAA), a leading non-governmental extension service provider in Nigeria with a strong focus on gender, ensures these tools address and integrate the needs of female farmers.

Pathway to Impact: Tailored fertilizer recommendations are expected to lead to 30–80% yield improvements. The advisory service aims to directly benefit 500,000 households and indirectly reach 1,000,000 households in key production areas of Nigeria. Read more

Call for Action

Implementing the proposed high-impact leverage points will bridge the gap between theory and practice, advancing soil health to improve agricultural productivity, environmental resilience, and inclusivity. Effective policy plays a vital role in promoting soil health through collaboration, R&D, knowledge exchange, and incentives. These policies must especially foster partnerships between farmers and industries, investment in infrastructure and enterprise development, prevent unwanted environmental risks and maximize social impact. Formulating agile regulations, smart subsidies, and effective incentives that consider result-based metrics are essential components of action plans.

There is consensus that emphasizing R&D will drive advancements in sustainable agronomic practices and fertilizer efficiency. While current policies have made progress, further development is needed to support long-term planning and risk management for farmers and landowners. Stakeholder advocacy for improved policies is limited by unclear political guidance. Therefore, policies should enhance coordination, innovation, and investment in soil health, secure land tenure, redesign subsidies, and provide targeted incentives aligned with market demand and sustainability. Our proposal aims to help stakeholders, including policymakers, prioritize actions, allocate resources, monitor progress, and maximize impact over time.

Conclusion

Improving soil health and fertilizer management is crucial for Africa's agricultural productivity and sustainability. This policy brief outlines practical steps to support the African Union's 10-year action plan for fertilizer and soil health. It highlights key issues and recommendations related to soil health, agronomy, and the challenges facing Africa's agricultural sector. Africa has vast arable land with significant agricultural potential but faces obstacles such as climate change, soil degradation, and socio-economic barriers that hinder sustainable productivity and food security.

Healthy soils are essential for sustainable agriculture, biodiversity conservation, and climate resilience. They support vigorous plant growth, efficient nutrient cycling, and water retention, which are critical for long-term agricultural viability. Aligning fertilizer use with soil health and environmental resilience is vital. The Africa Fertilizer and Soil Health Summit has emphasized the importance of integrating soil health with fertilizer use efficiency through multi-stakeholder interventions, robust R&D, and effective policy frameworks.

In this context, the CGIAR Excellence in Agronomy initiative is focused on scalable solutions to improve productivity, soil health, and climate resilience for smallholder farmers, especially women and youth. Addressing challenges like soil degradation, inadequate practices, and limited investment requires trusted partnerships, demand-driven R&D, targeted investments, and inclusive capacity development. Policy is essential for facilitating these efforts, driving soil health, agricultural sustainability, food security, and economic prosperity across the continent.

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